Input system:
Dependent variables:$x$
Independent variables:$t_2, t_1, t_3$
Equations:

$$\frac{\partial x}{\partial t_1} = x\sin(x)$$

$$\frac{\partial x}{\partial t_2} = x$$

$$\frac{\partial x}{\partial t_3} = 5$$

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Result:
Dependent variables:$s_1, s_2, x$
Independent variables:$t_2, t_1, t_3$
Additional variables:

$$s_1 = \sin(x)$$

$$s_2 = \cos(x)$$

Equations:

$$\frac{\partial x}{\partial t_1} = xs_1$$

$$\frac{\partial x}{\partial t_2} = x$$

$$\frac{\partial x}{\partial t_3} = 5$$

$$\frac{\partial s_1}{\partial t_2} = s_2 x$$

$$\frac{\partial s_1}{\partial t_1} = s_2 x s_1$$

$$\frac{\partial s_1}{\partial t_3} = 5s_2$$

$$\frac{\partial s_2}{\partial t_2} = (-1)\, s_1 x$$

$$\frac{\partial s_2}{\partial t_1} = (-1)\, s_1^2 x$$

$$\frac{\partial s_2}{\partial t_3} = (-5)\, s_1$$